

From White-Style Casimir Shells to Physical Readout Paths: High-Q, Superconducting, and Differential-Pressure Apparatus Options

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Abstract

White et al.'s sphere-in-cylinder Casimir calculation created a useful split in the experimental question. One part is geometric: whether a custom Casimir boundary condition can organize a shell-like negative vacuum-stress proxy near the region visually compared with an Alcubierre source shell. A recent role-resolved audit found that this shell-morphology layer survives several proxy and perturbation tests, while the direct metric/timing readout layer does not survive calibrated scale and electromagnetic-background accounting. The result is not a warp-readout program. It is a narrower physical program: identify whether a shell-coded Casimir boundary response can be transduced by ordinary precision apparatus strongly enough to be measured and separated from material, patch-potential, thermal, vibration, and readout backgrounds.

This paper starts from that narrowed claim boundary and translates the Stage 5 readout ladder into apparatus paths. The three remaining physical branches are a high-Q electromagnetic cavity frequency/phase shift, a superconducting resonator frequency/phase shift, and a paired differential-pressure cell. The first two are the strongest surviving branches under clean-control priors. The differential-pressure cell is more direct as force metrology but survives only under unusually strong patch-potential and common-mode control. A material or impedance readout is retained as a calibration/control branch, not as evidence for a shell response. The old central timing branch closes by amplitude.

The proposed continuation is therefore a Casimir-boundary metrology program. A positive result would not show practical warp behavior or a freely usable negative-energy fluid. It would show that a finite shell-coded boundary condition produces a repeatable renormalized-stress or boundary-mode response that survives physical controls in a modern apparatus. That would still be valuable: it would make the White-style geometry a precision test bed for finite-geometry Casimir physics, surface-patch systematics, high-Q resonant transduction, superconducting vacuum-boundary effects, and possible Casimir device engineering.

1. CLAIM BOUNDARY

White, Vera, Han, Brucoleri, and MacArthur used worldline numerics on custom Casimir geometries and reported that one sphere-in-cylinder configuration generated a negative energy-density morphology qualitatively similar to a two-dimensional Alcubierre shell [1]. The compact geometry is a $1\,\mu\text{m}$ diameter sphere centered in a nominal $4\,\mu\text{m}$ diameter cylinder. The important fact for this paper is not the visual resemblance alone. The useful fact is that the geometry separates three questions:

1. Does the boundary condition organize a shell-like Casimir response?
2. Is the calibrated stress-energy scale remotely competitive with a metric source requirement?
3. Is there a laboratory observable that can read the shell response through ordinary electromagnetic and material backgrounds?

The prior role-resolved audit answered these in different directions. The source-placement layer was positive in the limited proxy sense: shell morphology remained concentrated in the annular sphere-wall region and was controlled by the sphere and cylinder geometry. The metric-source and central timing layers were negative: the calibrated scale was many orders below an Alcubierre-type source demand, and the central readout path did not carry a recoverable physical signal. The synthetic discrimination layer added one more condition. If a shell-coded observable already reaches near-background amplitude, a geometry schedule can separate it from nuisance templates at about $\text{SBR} = 1.2$. The schedule is a discriminator, not an amplitude generator.

The Stage 5 run therefore changed the research question. The question is no longer whether the White geometry gives a direct warp or metric readout. In this harness, it does not. The question is whether the shell-coded Casimir boundary response can be coupled into a practical physical transducer strongly enough to pass an apparatus gate. The word “physical” is doing real work here: the branch must specify a sensor, a noise floor, dominant nuisance channels, calibration controls, a geometry modulation schedule, and a failure criterion.

2. NEGATIVE ENERGY IS NOT JUST A LABEL

The first conceptual question is whether these readouts would say anything about negative energy in quantum field theory, or whether “negative energy” is only bookkeeping. The careful answer is that QFT negative energy is operational but constrained. In renormalized stress-energy accounting, the local expectation value of energy density can be lower than the reference vacuum in some region. The Casimir effect is one familiar expression of this: boundaries alter the mode structure and produce measurable force, pressure, and energy changes. Ford’s review emphasizes both sides of the point: local negative energy densities are allowed in QFT, while quantum inequalities and compensating positive energy restrict their magnitude, duration, and use [3].

So a successful White-style physical readout would not merely attach a name to a calculation. If it survived geometry modulation, material substitution, surface-potential mapping, dummy-cell subtraction, and readout-nuisance controls, it would provide operational evidence that the finite boundary condition changed a measurable physical quantity in the direction predicted by renormalized vacuum-stress or Casimir-boundary theory. Depending on the branch, that quantity could be a force/pressure, a mechanical frequency shift, a cavity eigenfrequency shift, a microwave phase shift, or a superconducting resonator response.

But the same result would not elevate the system into an exotic-matter or warp-source device. The Stage 3 scale comparison already placed the gravitational/source magnitude far beneath representative metric demand. A positive high-Q or superconducting readout would mean: the shell-coded boundary condition is physically readable as a Casimir-boundary response. It would not mean: the negative energy has become an unconstrained material, extractable fuel, or engineered stress tensor suitable for macroscopic metric control. The strongest interpretation remains QFT-metrological, not propulsion-metrological.

3. STAGE 5 INPUTS

The Stage 5 priority physical run used the Stage 3 shell proxy and Stage 4 synthetic discrimination threshold as input. It evaluated five candidate readout families across physical backend cases. The run produced 33 candidate scenario gates, 27 physical-backend rows, 237600 synthetic observations, and

19800 fits. The declared gate threshold inherited from Stage 4 was $\text{SBR} = 1.2$, with ordinary-EM-only false-positive control near $\text{FP} < 0.05$.

Table 1: Stage 5 gate counts for the priority physical closeout run.

Gate status	Count
Candidate survives for experiment	11
Closed physical scale	14
Closed nuisance degeneracy	2
Closed amplitude	3
Background calibration only	3

Table 2: Highest-ranked Stage 5 survivors and near-survivors.

Branch	Scenario	Pred. SBR	FP	Dominant nuisance
High-Q cavity shift	optimistic clean	19.34	0.040	loss channel
Superconducting resonator	optimistic clean	17.76	0.0367	surface loss
Superconducting resonator	optimistic balanced	12.10	0.040	surface loss
High-Q cavity shift	optimistic matched	11.88	0.0267	loss channel
Superconducting resonator	optimistic lossy	7.94	0.0433	surface loss
High-Q cavity shift	optimistic lossy	6.07	0.0433	loss channel
Superconducting resonator	nominal clean	5.46	0.0267	surface loss
Differential pressure cell	optimistic quiet	5.31	0.0267	patch potential
Superconducting resonator	nominal balanced	3.59	0.0333	surface loss
High-Q cavity shift	nominal clean	2.22	0.0133	loss channel
Differential pressure cell	optimistic balanced	2.02	0.030	patch potential

The central timing reference is retained in the ladder as a closure check. Its optimistic physical row has predicted $\text{SBR} \sim 4.1 \times 10^{-59}$, and the three central timing rows all close by amplitude. The material impedance branch has predicted $\text{SBR} = 0$ by construction because it is a background and calibration channel, not a shell measurement. The surviving physical program is therefore concentrated in resonant boundary readouts, with differential pressure as a harder but conceptually direct force channel.

4. EXPERIMENTAL ARCHITECTURE

The viable apparatus cannot be a single static measurement. It has to be a geometry-modulated differential measurement, because the nuisance templates share many controls with the shell template. The experimental object is a finite boundary cell with controlled shell-gap modulation, sphere/cylinder geometry, wall thickness, readout coupling, and matched dummy geometry. A single absolute shift is not enough. The evidence is the vector of shifts across a modulation schedule.

The common architecture has five parts. First, a shell cell implements the sphere-cylinder or a manufacturable surrogate geometry with a known gap, surface roughness, conductivity, coating thickness, and alignment tolerance. Second, at least one dummy cell carries the same material, wiring, thermal mass, and readout chain while suppressing the shell overlap. Third, a geometry schedule changes the coordinates that the Stage 3/Stage 4 templates identified as discriminating: gap, cylinder radius, sphere size or surrogate radius, offset, wall thickness, length, and readout overlap. Fourth, an independent surface and material metrology layer maps roughness, residual potential, finite conductiv-

ity, coating thickness, loss tangent, and thermal expansion. Fifth, the readout is fitted against both a shell template and ordinary background templates, with an EM-only false-positive test.

This architecture is close to established Casimir metrology practice. Decca, Aksyuk, and Lopez review MEMS/NEMS force measurements as a natural platform for Casimir sensing [4]. Garcia-Sanchez et al. used metallized high-Q silicon nitride nanomembranes and in situ surface-potential measurement to control residual electrostatic forces [5, 6]. Bimonte et al. measured differential Casimir forces from 0.2 to 8 μm while explicitly treating surface roughness, edge effects, and patch potentials [9]. These are not optional niceties for the White-style follow-on; they are the minimum machinery required to keep a Casimir claim from becoming a material-background claim.

5. PATH A: HIGH-Q CAVITY SHIFT

The high-Q cavity branch treats the shell cell as a perturbation to an electromagnetic resonator. The observable is a fractional frequency or phase shift,

$$y_{\text{cav}} = \left(\frac{\delta f}{f} \right)_{\text{meas}}, \quad (1)$$

measured over a geometry schedule. In the Stage 5 physical backend, the best high-Q case used a shell fractional frequency shift of 9.40×10^{-11} against a background fractional rms of 4.86×10^{-12} , giving predicted $\text{SBR} = 19.34$. The strongest nominal clean case gave $\text{SBR} = 2.22$. Matched and lossy cases rapidly lose margin, which is exactly the point: the branch is viable only when the mode overlap and loss model are strong enough.

The physical device would use a cavity mode with high electric or magnetic field overlap in the shell-sensitive region and a companion null or low-overlap mode. A vector network analyzer, Pound-style lock, or phase-locked loop tracks the resonant frequency and quality factor. The fit model has the form

$$y_{\text{cav}}(q) = a_{\text{shell}}T_{\text{shell}}(q) + a_{\text{cond}}T_{\text{cond}}(q) + a_{\text{therm}}T_{\text{therm}}(q) + a_{\text{wg}}T_{\text{wg}}(q) + a_{\text{mix}}T_{\text{mix}}(q) + a_{\text{loss}}T_{\text{loss}}(q) + \epsilon, \quad (2)$$

where q denotes the geometry setting and the T_i are shell and nuisance templates. The cavity branch does not win by measuring a larger force directly. It wins by using resonance to make a small boundary-coded perturbation readable.

The required machinery is therefore mode-engineering machinery: a validated finite-element eigenmode solver; a measured mode-overlap factor; independent thermal expansion tracking; dummy cavities; controlled wall and conductor loss; and a way to reverse, detune, or suppress the shell-overlap mode while leaving the material and readout chain intact. Without these controls, a frequency shift is easy to obtain and hard to interpret. With them, a positive result would imply that the finite Casimir boundary condition couples to an electromagnetic eigenmode in a reproducible, geometry-coded way.

6. PATH B: SUPERCONDUCTING RESONATOR SHIFT

The superconducting resonator branch is the most interesting physically because it sits at the intersection of Casimir boundary conditions and cryogenic electrodynamics. The observable is again a fractional frequency or phase shift, but now the transducer is a superconducting resonator, microwave optomechanical structure, or superconducting boundary device. Stage 5 gave the best superconducting clean case $\text{SBR} = 17.76$, with nominal clean at $\text{SBR} = 5.46$ and nominal balanced at $\text{SBR} = 3.59$. The conservative and lossy rows show the main risk: surface loss, vortex loss, quasiparticle loss, kinetic inductance, and material-control channels can remain too close to the shell readout.

This branch has external motivation beyond the local harness. Perez et al. designed a NEMS system to probe Casimir energy corrections to superconducting condensation energy by measuring

superconducting transition changes in a controlled cavity [8]. de Jong et al. report measurement of the Casimir force between superconducting objects through the nonlinear dynamics of a superconducting drum resonator in a microwave optomechanical system [11]. Gurevich reviews the loss physics that must be confronted in superconducting resonators, including quasiparticles, dielectric loss, residual surface resistance, trapped vortices, and limits on high-Q operation [10]. These sources make the branch concrete: this is the kind of machine that can plausibly read a tiny boundary perturbation, and also the kind of machine whose internal loss channels can counterfeit one.

The clean experiment would use two resonator classes. The first is a material and loss reference resonator with the same film, substrate, wiring, field environment, and thermal cycle but no shell-sensitive Casimir geometry. The second is the shell-coupled resonator. The schedule should include temperature sweeps across and below T_c , controlled magnetic-field history for vortex loading, drive-power sweeps for quasiparticle and TLS response, and geometry modulation for shell template recovery. A strong positive result would have to remain locked to the shell geometry while changing differently from kinetic inductance, surface resistance, vortex, quasiparticle, and material-control templates.

If it worked, the physical implication would be narrower and sharper than a warp claim. It would indicate that superconducting boundary conditions or cryogenic resonant structures can transduce finite Casimir geometry changes into a reproducible microwave observable. Practical utility could include surface-loss diagnostics, Casimir-calibrated resonator sensors, vacuum-gap metrology, and tests of how superconducting electrodynamics enters finite geometry Casimir forces.

7. PATH C: DIFFERENTIAL PRESSURE CELL

The differential-pressure branch is conceptually the most direct. Instead of asking a resonator to magnify a boundary perturbation, it tries to measure a force or pressure contrast between matched cells. For ideal parallel plates, the pressure scale is

$$P_C(a) = -\frac{\pi^2 \hbar c}{240 a^4}, \quad (3)$$

which is about 1.3×10^{-3} Pa at $a = 1 \mu\text{m}$, before real geometry, material, finite-temperature, and roughness corrections. Stage 5's optimistic pressure rows sit in this same experimental neighborhood: the optimistic quiet row used a shell pressure of 7.18×10^{-4} Pa and passed with $\text{SBR} = 5.31$. The nominal quiet row fell to $\text{SBR} = 0.35$, and patchy or conservative rows close quickly.

The pressure branch therefore requires a paired-cell design rather than a single membrane. The two cells should share temperature, vibration, readout electronics, electrode materials, and membrane mechanics while moving through opposite shell-gap modulations. Patch-potential mapping is central, not auxiliary. Bimonte et al.'s long-range force measurement is directly relevant because it treats experimental uncertainty, patch potentials, roughness, edge effects, and theory comparison without fitting parameters [9]. Stange et al.'s commercial MEMS platform shows how a practical force sensor can be converted into a Casimir metrology platform with piconewton-scale resolution and electrostatic calibration [7]. The nanomembrane/Kelvin-probe work shows why surface-potential control must be built into the apparatus [5].

The branch's positive implication would be the cleanest statement of force physics: a shell-coded boundary geometry produces a pressure contrast that tracks the template after common-mode subtraction. Its risk is also the cleanest: patch potentials and ordinary electrostatic forces are naturally large compared with the desired residual. In the Stage 5 ladder, pressure is not dismissed, but it is the least forgiving survivor.

8. CONTROLS THAT ARE EVIDENCE-BEARING

The material impedance branch should be kept, but only as a control. It is a first-class competitor, not a shell readout. The most dangerous mistake would be to interpret any geometry-correlated impedance, capacitance, or loss change as evidence for a Casimir shell. Material response is exactly what must be measured so that it can be regressed out or used to close the claim.

Table 3: Control requirements by branch.

Branch	Required controls	Fail condition
High-Q cavity	Mode-overlap solver, dummy cavity, thermal monitor, conductor-loss model, waveguide/mode-mixing checks	Frequency shift follows loss, thermal, or waveguide template more strongly than shell template
Superconducting resonator	Material-only resonator, vortex history, quasiparticle and drive-power sweeps, surface-loss model, T_c sweep	Shift remains degenerate with kinetic inductance, surface loss, vortex, quasiparticle, or material-control channels
Differential pressure	Matched dummy cell, opposite gap modulation, Kelvin-probe map, roughness prior, vibration and thermal common-mode rejection	Residual pressure follows patch/electrostatic or membrane-drift template
Material impedance	Four-terminal calibration, capacitance/impedance mapping, dummy readout path	Treated as shell evidence rather than background calibration
Central timing	Matched central path and EM-only schedule	Closed by amplitude unless a new transduction model changes scale by many orders

The evidence-bearing form is a multi-template fit, not an isolated instrumental shift. A plausible shell readout must pass four conditions:

1. It exceeds the Stage 4 recovery threshold under the same geometry schedule, or a more stringent validated schedule.
2. EM-only and material-only synthetic or physical control datasets do not generate false positives above the declared rate.
3. A physically measured nuisance template is not more correlated with the data than the shell template.
4. The amplitude and sign remain stable under dummy-cell, material-swap, and geometry-reversal tests.

9. COMPARISON WITH WHITE ET AL.

The proposed program keeps White et al.’s most durable contribution and drops the fragile one. White et al. showed that a custom Casimir geometry can produce a shell-like vacuum-response morphology in a worldline calculation [1]. The prior audit did not erase that. It made the statement more precise: the shell morphology is a boundary-geometry result, while the metric-source and central-timing readouts fail under calibrated scale and background accounting.

The continuation proposed here changes the observable. White-style central or transit language asks whether the apparatus creates a direct metric-like effect readable along the center. The Stage 5 answer is no for the modeled apparatus. The high-Q, superconducting, and pressure branches instead

ask whether the same shell-coded boundary geometry can be read through ordinary Casimir metrology. That is a much more conservative question, but it is also physically buildable.

The distinction matters for interpretation. A positive resonator or pressure result would strengthen the claim that the finite shell geometry is a real physical Casimir boundary condition, not a plotting artifact. It would not resurrect the direct warp-readout interpretation. The best narrative is: White et al. identified a provocative shell-forming Casimir geometry; the audit separated morphology from metric source scale; Stage 5 identified the physical readout branches worth testing as Casimir-boundary metrology.

10. BUILD PLAN

The practical next program should be staged so that each step can close the line cleanly.

10.1 Phase 0: apparatus digital twin

Build branch-specific simulations before fabricating the shell cell. For the cavity branch, solve eigenmodes and perturbative frequency shifts for the actual geometry, including conductor thickness, ports, readout feed, and wall loss. For superconducting resonators, include kinetic inductance, surface resistance, quasiparticle, vortex, TLS, and material-control terms. For pressure cells, include membrane mechanics, patch maps, roughness, electrostatic calibration, and edge corrections. The output of Phase 0 is a measured or simulated template matrix that can be passed through the Stage 4/Stage 5 synthetic gates.

10.2 Phase 1: known-geometry calibration

Do not begin with the White shell cell. First reproduce a standard sphere-plane, parallel-plate, trench, or membrane Casimir measurement at the chosen scale. The purpose is not novelty. It is to prove the apparatus can recover an ordinary Casimir signal while controlling residual electrostatics, roughness, thermal drift, and readout artifacts. This phase should use the Decca/Bimonte/Garcia-Sanchez style controls: electrostatic calibration, Kelvin-probe surface-potential mapping where possible, separation metrology, and explicit error budgets [4, 9, 5].

10.3 Phase 2: dummy and material controls

Construct dummy geometries with the same materials and readout chain but suppressed shell overlap. Run the full geometry schedule on shell and dummy cells. This is where the material impedance branch earns its keep. A large geometry-correlated material response is not a surprise; it is the main background map.

10.4 Phase 3: shell-cell measurement

Run the shell cell under the full schedule. The acceptance gate should be predeclared: predicted and measured SBR, false-positive rate, nuisance correlations, mode/loss separation, and branch-specific amplitude bounds. If the high-Q and superconducting branches fail after measured template updates, the line should close as a physical readout program. If one branch survives, the next paper should be an apparatus design and uncertainty-budget paper, not a metric interpretation paper.

11. PHYSICAL IMPLICATIONS AND UTILITY

If all three branches fail under measured apparatus templates, the physical conclusion is still useful. It would say that the White-style shell morphology is a real proxy-level organization of Casimir boundary

response, but that available transducers do not lift it through ordinary material and electromagnetic backgrounds. That would close the current line at the metrology gate.

If the pressure branch survives, the implication is a direct finite-geometry force result. It would show that a shell-coded Casimir boundary condition can produce a pressure contrast that remains after patch and common-mode controls. Practical utility would be strongest in precision force metrology, MEMS/NEMS calibration, and surface-patch diagnostics.

If the high-Q cavity branch survives, the implication is resonant electromagnetic transduction. The shell boundary condition would be readable as a mode shift rather than as a direct force. Practical utility could include Casimir-assisted cavity sensors, mode-overlap diagnostics for microcavities, and precision tests of finite-geometry vacuum-boundary calculations.

If the superconducting branch survives, the implication is the most interesting for condensed-matter/QFT overlap. It would suggest that superconducting boundary conditions, resonator loss physics, and Casimir geometry can be separated well enough to measure a shell-coded vacuum-boundary effect. Practical utility could include cryogenic gap metrology, superconducting surface-loss diagnostics, and improved understanding of how superconducting electrodynamics enters Casimir forces.

None of these outcomes would imply practical warp propulsion. The value is smaller but real: a controlled way to ask how engineered boundaries shape renormalized vacuum stress and how those changes can be read by precision machines.

12. CODE, DATA, AND SOURCE AVAILABILITY

The local source bundle for this paper is stored at

```
white_casimir_intersection/sources
```

and is documented in

```
white_casimir_intersection/sources/README.md.
```

The Stage 5 priority physical closeout used here is stored under:

```
/media/kir/9CDCBD3EDCBD140C/Research/  
white_casimir_stage5_readout_ladder/stage5_priority_physical_closeout_20260530
```

The compact files used for the tables and claims are:

- `summary.json`
- `gates/gate_summary.csv`
- `candidates/readout_candidates.csv`
- `backends/physical_backend_sweep.csv`

13. CONCLUSION

The current state of the White Casimir line is narrow but not empty. The central timing and metric-readout interpretation is closed in the present harness. The shell morphology is still useful as a finite Casimir boundary configuration. Stage 5 says that the strongest physical continuations are resonant readouts: a high-Q cavity and a superconducting resonator. A differential-pressure cell remains a possible force readout only under strong patch and common-mode control.

The next honest question is whether one of those physical branches can survive measured apparatus templates. If the answer is yes, the result would be a valuable Casimir-boundary readout, not a warp signal. If the answer is no, the research line closes cleanly: White-style shell morphology exists at the proxy level, but no physically robust readout remains after ordinary backgrounds and controls.

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